



Field and Wave Electromagnetics

电磁场与电磁波

Chapter 5: Steady Electric Currents

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Chapter Title		课时	
1	Static Electric Fields	3	静电场： (1) 期末<15%，不考大题； (2) 磁矢等难度大尽量简化
2	Solution of Electrostatic Problems		
4	Steady Electric Currents	2	
5	Basic Static Magnetic Fields	2	
7	Time-varying Fields and Maxwell's Equations	5	
9	Plane Electromagnetic Waves	11	电压计算等适当简化，
13	Waveguides and Cavity Resonators	11	重点强化章节，期末有一道推导大题
16	Antennas and Radiating systems	3	
	机动/答疑	2	
		40	

1 总体原则：弱化电路相关(V,A,I,U)，强化Photonics相关(E,H)，突出本课程对光电专业支撑作用

2 删掉“传输线”，增加“天线”；删除“镜像法”（分离变量法保留）

3 课时可以根据情况适当调整，预留2课时为机动/答疑课时；

4 平时成绩40分，（作业上交10分+点名10分，作业完成情况20分）

5 作业完成情况20分学生两种选择（默认选择a）：

（a）作业上交点名20分+作业完成情况(绑定期末作业题)20分

（b）助教随机抽出4个作业大题，每题5分，1小时完成(答疑课时候完成)，得分视为平时成绩。

Chapter 3: Static Electric Fields

Field: A spatial distribution of a scalar or vector quantity which may or may not be a function of time.

1. Fundamental postulates of electrostatic in free space

2. Coulomb's law

3. Gauss's law

4. Conductors in static electric field

5. Dielectrics in static electric field

6. Electric flux density

7. Boundary conditions for electrostatic fields

8. Capacitance and capacitors

9. Electrostatic energy and forces

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0} \quad \oint_S \vec{E} \cdot d\vec{s} = \frac{Q}{\epsilon_0}$$

$$\nabla \times \vec{E} = 0 \quad \oint_C \vec{E} \cdot d\vec{l} = 0$$

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P}$$

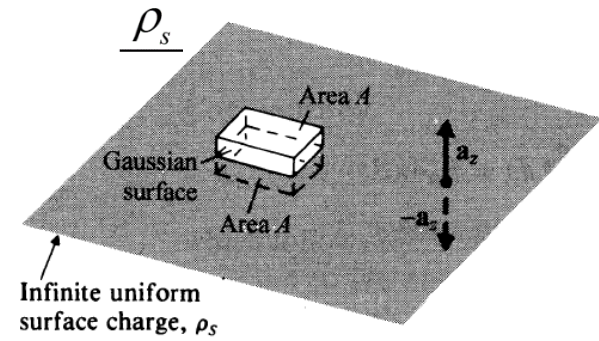
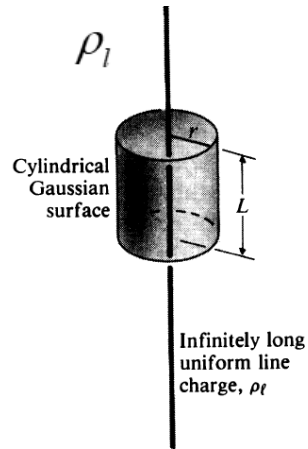
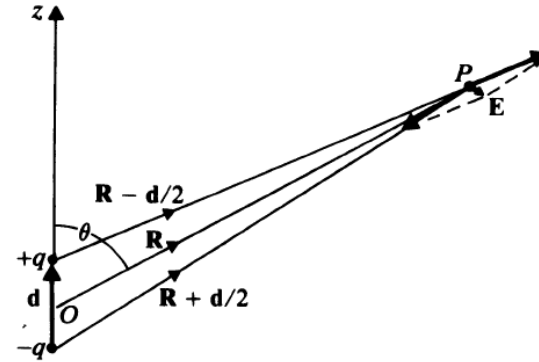
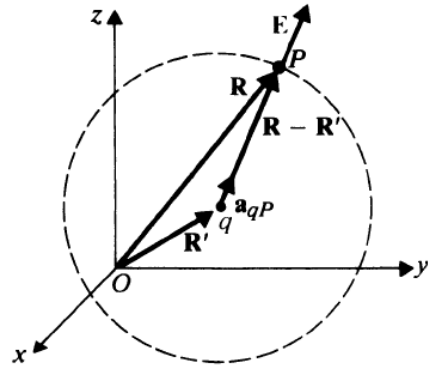
$$\nabla \cdot \vec{D} = \rho$$

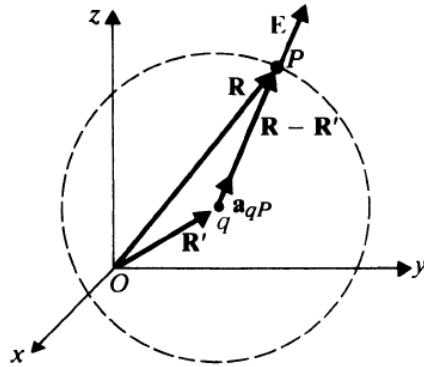
$$\oint_S \vec{D} \cdot d\vec{s} = Q$$

Two fundamental postulates

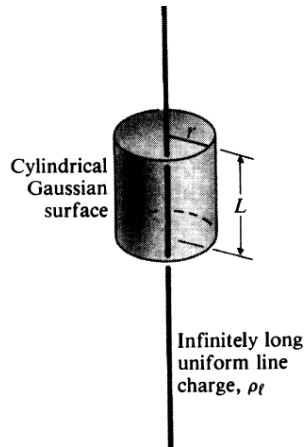
$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0} \quad \oint_S \vec{E} \cdot d\vec{s} = \frac{Q}{\epsilon_0}$$

$$\nabla \times \vec{E} = 0 \quad \oint_C \vec{E} \cdot d\vec{l} = 0$$

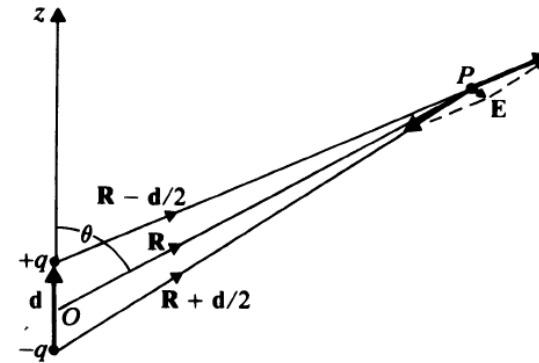




$$\vec{E}_p = \frac{q(\vec{R} - \vec{R}')}{4\pi\epsilon_0 |\vec{R} - \vec{R}'|^3}$$



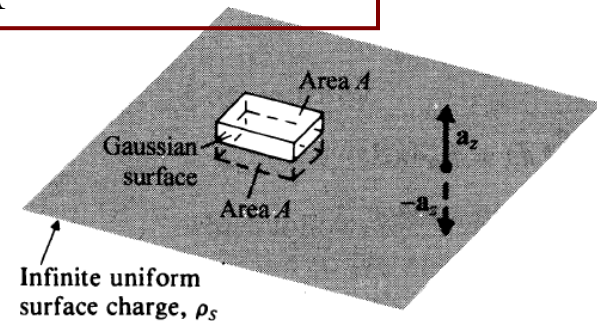
$$\vec{E} = \vec{a}_r E_r = \vec{a}_r \frac{\rho_l}{2\pi\epsilon_0 r}$$



$$\vec{E} = \frac{1}{4\pi\epsilon_0 R^3} \left(\frac{\vec{R} \times \vec{p}}{R^2} \times \vec{R} - \vec{p} \right)$$

$$\vec{p} = q\vec{d}$$

$$\vec{E} = \frac{p}{4\pi\epsilon_0 R^3} (\vec{a}_R 2 \cos \theta + \vec{a}_\theta \sin \theta)$$



$$\vec{E} = \vec{a}_z E_z = \vec{a}_z \frac{\rho_s}{2\epsilon_0}, \quad (z > 0)$$

$$\vec{E} = -\vec{a}_z E_z = -\vec{a}_z \frac{\rho_s}{2\epsilon_0}, \quad (z < 0)$$

Electric Potential and Dielectrics

$$\boxed{\vec{E} = -\nabla V} \Leftrightarrow \begin{cases} \nabla \times \vec{E} = 0 \\ \nabla \times (\nabla A) = 0 \end{cases}$$

因为 $\mathbf{E}$ 的旋度为零，这意味着 $\mathbf{E}$ 必然可以表示为一个标量的梯度
(因为梯度的旋度恒为零)

$$\boxed{V_2 - V_1 = - \oint_{P_1}^{P_2} \vec{E} \cdot d\vec{l}}$$

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P}$$

$$\nabla \cdot \vec{D} = \rho$$

$$\oint_S \vec{D} \cdot d\vec{s} = Q$$

Maxwell's Equations

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

$$\nabla \cdot \vec{D} = \rho$$

$$\nabla \cdot \vec{B} = 0$$

Chapter 4: Solutions of Electrostatic Problems

4-1 Introduction

4-2 Poisson's and Laplace's Equations $\nabla^2 V = -\frac{\rho}{\epsilon}$ $\nabla^2 V = 0$

4-3 Uniqueness of Electrostatic Solutions

4-4 Methods of Images

4-5 Boundary-Value Problems in Cartesian Coordinates

4-6 Boundary-Value Problems in Cylindrical Coordinates

4-7 Boundary-Value Problems in Spherical Coordinates

4.2 Poisson's and Laplace's Equations

Maxwell Equation:

$$\nabla \cdot \vec{D} = \rho \quad \text{free charges}$$

$$\nabla \times \vec{E} = 0$$

$$\vec{E} = -\nabla V$$

In a linear and isotropic medium: $\vec{D} = \epsilon \vec{E}$

$$\nabla \cdot \epsilon \vec{E} = \rho$$

$$\nabla \cdot (\epsilon \nabla V) = -\rho$$

For a homogeneous medium:

$$\nabla^2 V = -\frac{\rho}{\epsilon}$$

Poisson's equation:

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = -\frac{\rho}{\epsilon}$$

$$\nabla^2 V = \nabla \cdot \nabla V = \left(\vec{a}_x \frac{\partial}{\partial x} + \vec{a}_y \frac{\partial}{\partial y} + \vec{a}_z \frac{\partial}{\partial z} \right) \cdot \left(\vec{a}_x \frac{\partial V}{\partial x} + \vec{a}_y \frac{\partial V}{\partial y} + \vec{a}_z \frac{\partial V}{\partial z} \right)$$

$$= \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

4.2 Poisson's and Laplace's Equations

Cylindrical coordinates:

$$\nabla^2 V = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial V}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 V}{\partial \Phi^2} + \frac{\partial^2 V}{\partial z^2}$$

Spherical coordinates:

$$\nabla^2 V = \frac{1}{R^2} \frac{\partial}{\partial R} \left(R^2 \frac{\partial V}{\partial R} \right) + \frac{1}{R^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) + \frac{1}{R^2 \sin^2 \theta} \frac{\partial^2 V}{\partial \Phi^2}$$

no free charge Laplace's equation

$$\nabla^2 V = 0$$

electric field

$$\vec{E} = -\nabla V$$

**charge distribution on the
conductor surfaces**

$$\rho_s = \epsilon \vec{E}_n$$

Determine the field both inside and outside a spherical cloud of electrons with a uniform volume charge density $\rho = -\rho_0$ (where ρ_0 is a positive quantity) for $0 \leq R \leq b$ and $\rho = 0$ for $R > b$ by solving Poisson's and Laplace's equation for V .

(1) Inside the cloud $0 \leq R \leq b$ $\rho = -\rho_0$

Poisson's equation:

$$\nabla^2 V = -\frac{\rho}{\epsilon}$$

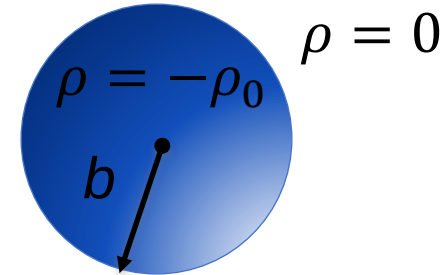
$$\frac{1}{R^2} \frac{d}{dR} \left(R^2 \frac{dV_i}{dR} \right) = \frac{\rho_0}{\epsilon_0} \qquad \frac{d}{dR} \left(R^2 \frac{dV_i}{dR} \right) = \frac{\rho_0}{\epsilon_0} R^2$$

$$\frac{dV_i}{dR} = \frac{\rho_0}{3\epsilon_0} R + \frac{C_1}{R^2}.$$

The electric field intensity inside

$$\vec{E}_i = -\nabla V_i = -\vec{a}_R \left(\frac{dV_i}{dR} \right)$$

$$\vec{E}_i = -\vec{a}_R \left(\frac{\rho_0}{3\epsilon_0} \right) R$$



$$\nabla^2 V = \frac{1}{R^2} \frac{\partial}{\partial R} \left(R^2 \frac{\partial V}{\partial R} \right) + \frac{1}{R^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) + \frac{1}{R^2 \sin^2 \theta} \frac{\partial^2 V}{\partial \Phi^2}$$

Determine the field both inside and outside a spherical cloud of electrons with a uniform volume charge density $\rho = -\rho_0$ (where ρ_0 is a positive quantity) for $0 \leq R \leq b$ and $\rho = 0$ for $R > b$ by solving Poisson's and Laplace's equation for V .

(2) Outside the cloud $b \leq R$ $\rho = 0$

Laplace's equation $\nabla^2 V = 0$

$$\frac{1}{R^2} \frac{d}{dR} \left(R^2 \frac{dV_o}{dR} \right) = 0$$

$$\frac{dV_o}{dR} = \frac{C_2}{R^2}$$

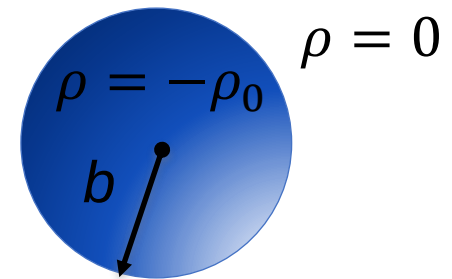
The electric field $\vec{E}_0 = -\nabla V_0 = -\vec{a}_R \left(\frac{dV_0}{dR} \right) = -\vec{a}_R \left(\frac{C_2}{R^2} \right)$

The electric field continuity at $R=b$ $\frac{C_2}{b^2} = \frac{\rho_0}{3\epsilon_0} b$

$$\vec{E}_0 = -\vec{a}_R \frac{\rho_0 b^3}{3\epsilon_0 R^2}$$

$$Q = -\rho_0 \frac{4\pi}{3} b^3$$

$$\vec{E}_0 = \vec{a}_R \frac{Q}{4\pi\epsilon_0 R^2}$$



inside $\frac{dV_i}{dR} = \frac{\rho_0}{3\epsilon_0} R + \frac{C_1}{R^2}$

4.3 Uniqueness of Electrostatic Solutions

Uniqueness theorem

A solution of Poisson's equation (of which Laplace's equation is a special case) that satisfies the given boundary conditions is *a unique solution*.

A solution of an electrostatic problem satisfying its boundary conditions is the only possible solution, *irrespective of the method by which the solution is obtained*.

4.5 Boundary-value problems in Cartesian Coordinates

Potential equation (free source):

$$\nabla^2 V = 0$$

Method of images: free charges near conducting boundaries.

For problems consisting of a system of conductors maintained at specified V and with no isolated free charges, how to solve them?

Method of separation of variables: the solution can be expressed as a product of three one-dimensional functions, each depending separately on one coordinate variable only

Three types of boundary-value problem:

- (1) V is specified;
- (2) the normal derivative of V is specified;
- (3) V or the normal derivative of V is specified;

4.5 Boundary-value problems in Cartesian Coordinates

Laplace's equation for V :

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0$$

$$V(x, y, z) = X(x)Y(y)Z(z)$$

$X(x), Y(y)$ and $Z(z)$ are functions of only x, y and z .

$$Y(y)Z(z)\frac{d^2 X(x)}{dx^2} + X(x)Z(z)\frac{d^2 Y(y)}{dy^2} + X(x)Y(y)\frac{d^2 Z(z)}{dz^2} = 0$$

$$\frac{1}{X(x)}\frac{d^2 X(x)}{dx^2} + \frac{1}{Y(y)}\frac{d^2 Y(y)}{dy^2} + \frac{1}{Z(z)}\frac{d^2 Z(z)}{dz^2} = 0$$

In order for Eq. to be satisfied for all values of x, y, z , each of the three terms must be a constant.

4.5 Boundary-value problems in Cartesian Coordinates

$$\frac{d}{dx} \left[\frac{1}{X(x)} \frac{d^2 X(x)}{dx^2} \right] = 0$$

$$\frac{1}{X(x)} \frac{d^2 X(x)}{dx^2} = -k_x^2$$

k_x^2 is a constant to be determined from the boundary conditions
 k_x is imaginary, $-k_x^2$ is a positive real number
 k_x is real, $-k_x^2$ is a negative real number

$$\frac{d^2 X(x)}{dx^2} + k_x^2 X(x) = 0$$

$$\frac{d^2 Y(y)}{dy^2} + k_y^2 Y(y) = 0 \quad k_x^2 + k_y^2 + k_z^2 = 0$$

$$\frac{d^2 Z(z)}{dz^2} + k_z^2 Z(z) = 0$$

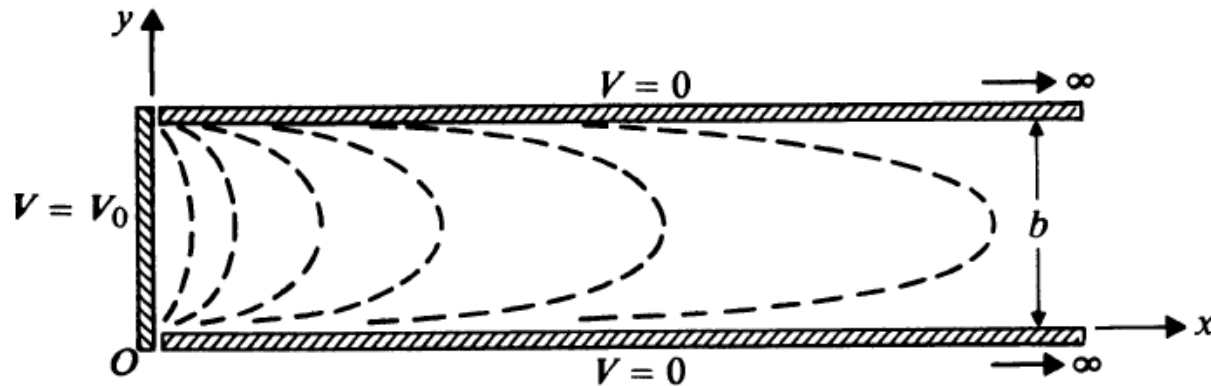
4.5 Boundary-value problems in Cartesian Coordinates

$$\frac{d^2 X(x)}{dx^2} + k_x^2 X(x) = 0$$

k_x^2	k_x	$X(x)$
0	0	$A_0 x + B_0$
+	k	$A_1 \sin kx + B_1 \cos kx$
-	jk	$A_2 \sinh kx + B_2 \cosh kx$

4.5 Boundary-value problems in Cartesian Coordinates

Two grounded, semi-infinite, parallel-plane electrodes are separated by a distance b . A third electrode perpendicular to and insulated from both is maintained at a constant. Determine the potential distribution in the region enclosed by the electrodes.



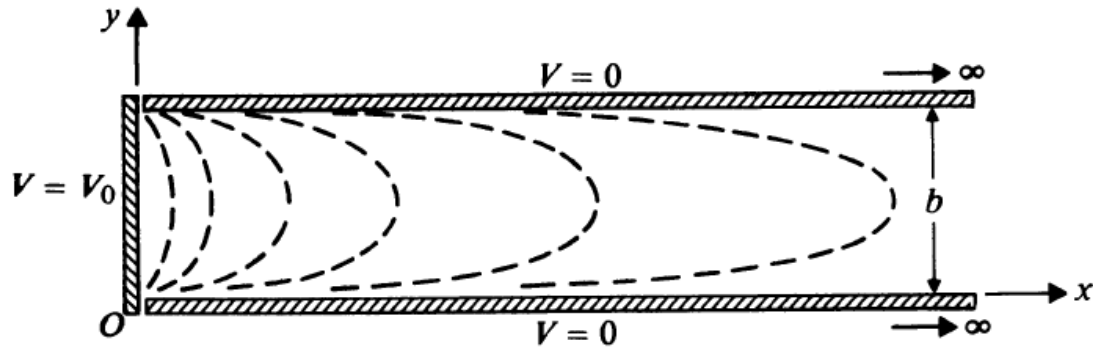
With V independent of z $V(x, y, z) = V(x, y)$

In the x direction $V(0, y) = V_0$ $V(\infty, y) = 0$

In the y direction $V(x, 0) = 0$ $V(x, b) = 0$

$$k_z = 0 \qquad Z(z) = B_0$$

4.5 Boundary-value problems in Cartesian Coordinates



In the x direction $V(0, y) = V_0$ $V(\infty, y) = 0$

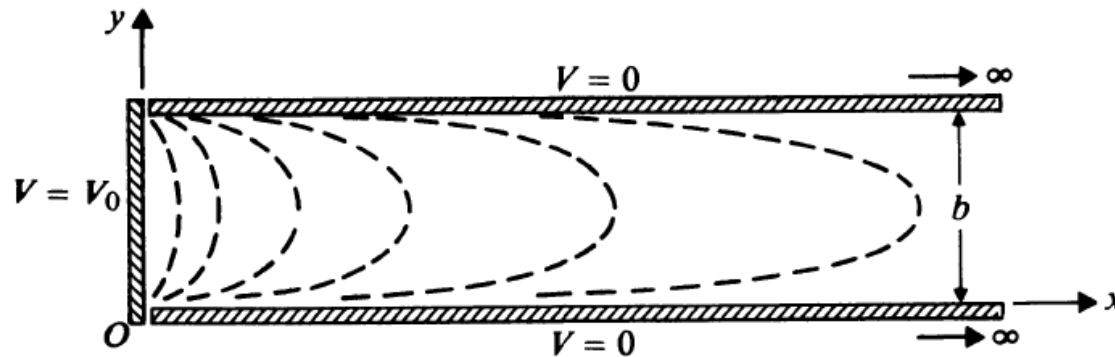
In the y direction $V(x, 0) = 0$ $V(x, b) = 0$

$$X(x) = D_2 e^{-kx}$$

$$Y(y) = A_1 \sin ky$$

$$V_n(x, y) = (B_0 D_2 A_1) e^{-kx} \sin ky = C_n e^{-kx} \sin ky$$

4.5 Boundary-value problems in Cartesian Coordinates



$$V_n(x, b) = C_n e^{-kx} \sin kb = 0$$

Should be satisfied for all values of x , only if

$$k = \frac{n\pi}{b}, n = 1, 2, 3, \dots$$

$$V_n(x, y) = C_n e^{-\frac{n\pi}{b}x} \sin \frac{n\pi}{b} y$$

$$V(0, y) = \sum_{n=1}^{\infty} V_n(0, y) = \sum_{n=1}^{\infty} C_n \sin \frac{n\pi}{b} y = V_0$$

$$C_n = \begin{cases} \frac{4V_0}{n\pi}, & n - \text{odd} \\ 0, & n - \text{even} \end{cases}$$

Solution of Electrostatic Problems

Method of images: free charges near conducting boundaries.

Method of separation of variables: for problems consisting of a system of conductors maintained at specified potentials and with no isolated free charges.

Method of separation of variables: the solution can be expressed as a product of three one-dimensional functions, each depending separately on one coordinate variable only

Chapter 5: Steady Electric Currents

5-1 Introduction

5-2 Current density and Ohm's law

5-3 Electromotive force and Kirchhoff's voltage law

5-4 Continuity of equation and Kirchhoff's current law

5-5 Power dissipation and Joule's law

5-6 Boundary conditions for current density

5-7 Resistance calculation

5-1 Introduction

Types of electric currents:

■ Conduction current 传导电流:

Drift motion of conduction **electrons and/or holes** in **conductors and semiconductors**; very low for the average drift velocity of the electrons (1/1000 m/s);

■ Electrolytic current 电解电流:

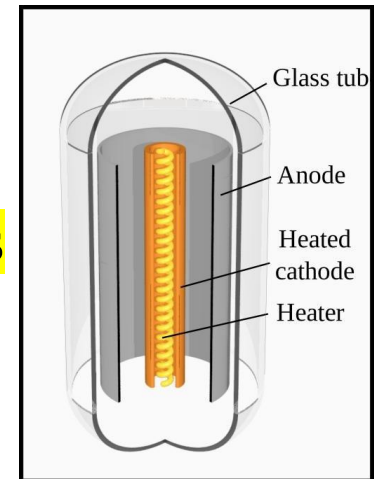
Migration of positive and negative **ions**;

■ Convection current 对流电流 :

Motion of electrons and/or ions **in vacuum or in rarefied gas** not governed by Ohm's law.

■ Displacement current 位移电流 :

The time-rate of change of electric displacement dD/dt



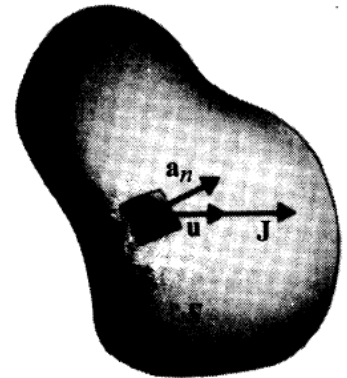
5-2 Current density and Ohm's law

Consider a steady motion of one kind of charges q over a differential surface Δs with velocity $\mathbf{u}$ and the total charges ΔQ passing this surface in a time interval Δt is N is the number of charge carriers per unit volume

$$\Delta Q = Nq\mathbf{u} \cdot \mathbf{a}_n \Delta s \Delta t$$

Total current over the differential surface Δs :

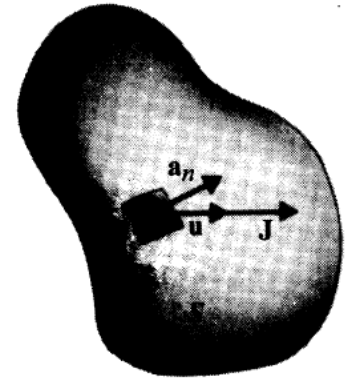
$$\Delta I = \frac{\Delta Q}{\Delta t} = Nq\mathbf{u} \cdot \mathbf{a}_n \Delta s = \underline{Nq\mathbf{u} \cdot \Delta \mathbf{s}}$$



Current density: $\mathbf{J} = Nq\mathbf{u} \quad (\text{A/m}^2) \quad \longrightarrow \quad \Delta I = \mathbf{J} \cdot \Delta \mathbf{s}.$

The total current I flowing through an arbitrary surface S :

$$I = \int_S \mathbf{J} \cdot d\mathbf{s} \quad (\text{A}).$$



Noting that the charge density $\rho = Nq$, we have

$$\mathbf{J} = \rho \mathbf{u} \quad (\text{A/m}^2), \quad \longleftarrow \quad \mathbf{J} = Nq\mathbf{u}$$

Which is the relation between the **convection current density** and the **velocity** of the charge carrier

In the case of conduction currents there may be more than one kind of charge carriers (electrons, holes, and ions) drifting with different velocities. Equation (5-3) should be generalized to read

$$\mathbf{J} = \sum_i N_i q_i \mathbf{u}_i \quad (\text{A/m}^2).$$

- Drift motion of charge carriers under electric field
- Atoms remain neutral ($\rho=0$)

For most of metallic conductors in which the average electron drift is proportional to electric field, we write

$$\mathbf{u} = -\mu_e \mathbf{E} \quad (\text{m/s}),$$

Electron mobility μ_e :

$$\text{Cu: } 3.2 \times 10^{-3} \text{ (m}^2/\text{V} \cdot \text{s)}.$$

$$\text{Al: } 1.4 \times 10^{-4} \text{ (m}^2/\text{V} \cdot \text{s)}$$

$$\mathbf{J} = -\rho_e \mu_e \mathbf{E}, \quad \begin{array}{l} \leftarrow \mathbf{u} = -\mu_e \mathbf{E} \\ \leftarrow \mathbf{J} = \rho \mathbf{u} \end{array}$$

$\rho_e = -Ne$: charge density of the drifting electrons.

$$\boxed{\mathbf{J} = \sigma \mathbf{E} \quad (\text{A/m}^2),}$$

Conductivity, $\sigma = -\rho_e \mu_e$
a macroscopic parameter of a medium

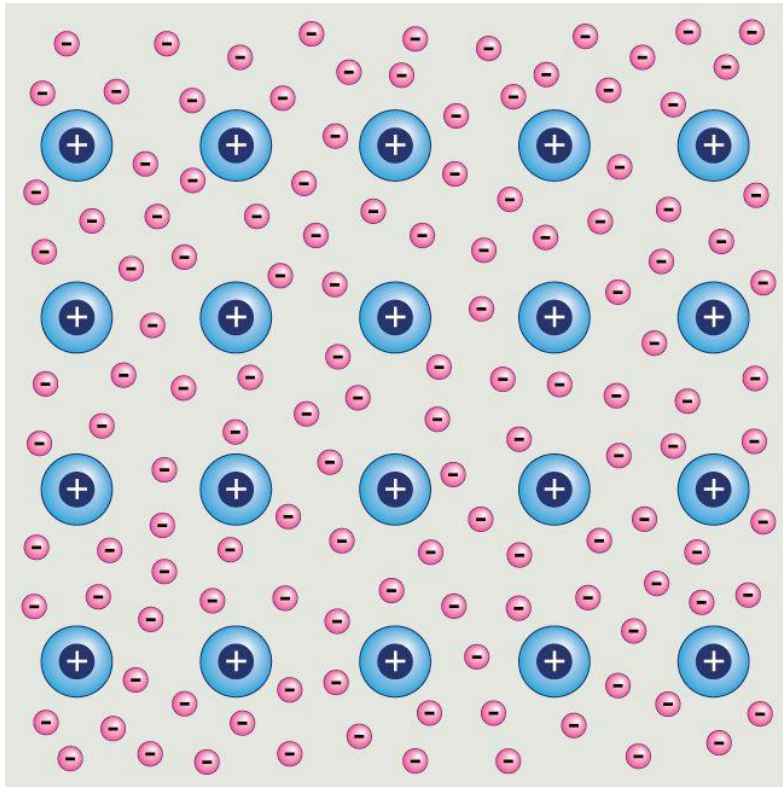
For semiconductors, conductivity depends on the concentration and mobility of both electrons and holes:

$$\sigma = -\rho_e \mu_e + \rho_h \mu_h, \quad \text{Graphene: } 1.5 \text{ (m}^2/\text{V s)} \quad (5-22)$$

where the subscript h denotes hole. In general, $\mu_e \neq \mu_h$. For germanium, typical values are $\mu_e = 0.38$, $\mu_h = 0.18$; for silicon, $\mu_e = 0.12$, $\mu_h = 0.03$ (m²/V·s).

Drude model: conductivity for metals (Microscopic Ohm's law)

Metal and Free electron gas



Nuclei fixed; electron are free moving

Harmonic oscillation of electrons under the excitation of electric fields

$$m_e \vec{a}_e = \vec{F}_{E_local} + \vec{F}_{damping}$$

$$m_e \frac{d\vec{u}}{dt} = -e\vec{E}_L e^{-i\omega t} - m_e \underset{\substack{\uparrow \\ \text{Collision frequency}}}{g} \vec{u}$$

$$\Rightarrow \vec{u} = \frac{e}{im_e \omega - m_e g} \vec{E}_L e^{-i\omega t}$$

$$\Rightarrow \vec{J} = -Ne\vec{u} = \left(\frac{-Ne^2}{im_e \omega - m_e g} \right) \vec{E}_L e^{-i\omega t}$$

$$\Rightarrow s(\omega) = \frac{Ne^2}{-im_e \omega + m_e g} = \frac{Ne^2}{m_e g} \frac{1}{(1 - i\omega/g)}$$

Conductivity (static)

$$s(0) = \frac{Ne^2}{m_e g}$$

Material	Conductivity, $\sigma(\text{S/m})$	Material	Conductivity, $\sigma(\text{S/m})$
Silver	6.17×10^7	Fresh water	10^{-3}
Copper	5.80×10^7	Distilled water	2×10^{-4}
Gold	4.10×10^7	Dry soil	10^{-5}
Aluminum	3.54×10^7	Transformer oil	10^{-11}
Brass	1.57×10^7	Glass	10^{-12}
Bronze	10^7	Porcelain	2×10^{-13}
Iron	10^7	Rubber	10^{-15}
Seawater	4	Fused quartz	10^{-17}

Resistivity: the reciprocal of conductivity ($\Omega \cdot \text{m}$)

Resistance and conductance

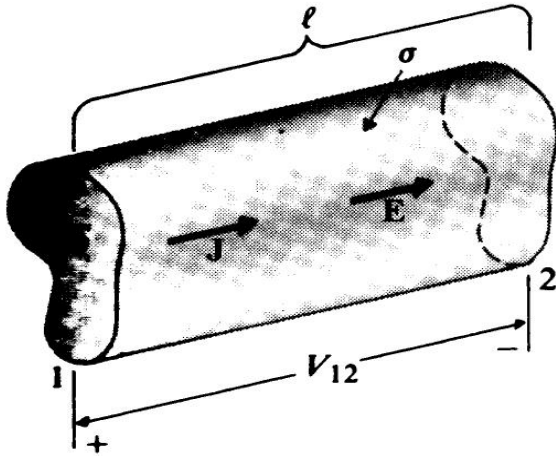


Fig. 5-3 Homogeneous conductor with a constant cross section

Voltage between terminals 1 and 2:

$$V_{12} = E\ell$$

The total current:

$$I = \int_S \mathbf{J} \cdot d\mathbf{s} = JS$$

Point or microscopic form of Ohm's law

$$\mathbf{J} = \sigma \mathbf{E}$$

$$\frac{I}{S} = \sigma \frac{V_{12}}{\ell}$$

Resistance

$$R = \frac{\ell}{\sigma S}$$

$$V_{12} = \left(\frac{\ell}{\sigma S} \right) I = RI$$

Macroscopic Ohm's law

Resistance and conductance

The **conductance**, G , or the reciprocal of resistance, is useful in combining resistances in parallel. The unit for conductance is (Ω^{-1}), or siemens (S).

$$G = \frac{1}{R} = \sigma \frac{S}{\ell} \quad (\text{S}). \quad (5-28)$$

From circuit theory we know the following:

- a)** When resistances R_1 and R_2 are connected in series (same current), the total resistance R is

$$\boxed{R_{sr} = R_1 + R_2.} \quad (5-29)$$

- b)** When resistances R_1 and R_2 are connected in parallel (same voltage), we have

$$\boxed{\frac{1}{R_{||}} = \frac{1}{R_1} + \frac{1}{R_2}} \quad (5-30a)$$

高温超导材料的简介

1911年荷兰物理学家发现电阻完全消失的现象仅发生在物质处于极端物理状态的温度-临界温度。

1933年，德国科学家发现超导体的抗磁效应。

1962年，英国剑桥大学博士研究生证明两个以薄的绝缘层相隔的超导体之间会产生一种特殊现象-隧道效应。

1962年，美国Westinghose公司用钛化铌合金拉制成超导线。 (15-25K)

1986年，在苏黎世附近一家IBM实验室工作的两位瑞士物理学家宣布，他们研制成含镧和钡的铜酸盐（氧化铜）化合物，其临界温度为30K。更令人惊奇的是此物质为一种陶瓷材料，在常温下具有绝缘体的所用特征。

在随之而来的高温超导热中又发现了一批具有较高临界温度的化合物，目前临界温度的最高纪录约是150K (-123°C)

室温超导前沿研究

Article | [Published: 08 March 2023](#)

RETRACTED ARTICLE: Evidence of near-ambient superconductivity in a N-doped lutetium hydride

[Nathan Dasenbrock-Gammon](#), [Elliot Snider](#), [Raymond McBride](#), [Hiranya Pasan](#), [Dylan Durkee](#), [Nugzari Khelvashi-Sutter](#), [Sasanka Munasinghe](#), [Sachith E. Dissanayake](#), [Keith V. Lawler](#), [Ashkan Salamat](#) & [Ranga P. Dias](#) 

[Nature](#) **615**, 244–250 (2023) | [Cite this article](#)

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 This article was [retracted](#) on 07 November 2023

This article is in the 99th percentile (ranked 77th) of the 232,594 tracked articles of a similar age in all journals and the 99th percentile (ranked 6th) of the 562 tracked articles of a similar age in *Nature*

合物能以零电阻的形式传导
电流。

室温超导前沿研究

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Article | [Published: 14 October 2020](#)

RETRACTED ARTICLE: Room-temperature superconductivity in a carbonaceous sulfur hydride

[Elliot Snider](#), [Nathan Dasenbrock-Gammon](#), [Raymond McBride](#), [Mathew Debessai](#), [Hiranya Vindana](#), [Kevin Vencatasamy](#), [Keith V. Lawler](#), [Ashkan Salamat](#) & [Ranga P. Dias](#) 

[Nature](#) **586**, 373–377 (2020) | [Cite this article](#)

128k Accesses | 405 Citations | 3141 Altmetric | [Metrics](#)

 This article was [retracted](#) on 26 September 2022

 **15 February 2022** Editor's Note: The editors of Nature have been alerted to concerns regarding the manner in which the data in this paper have been processed and interpreted. Nature is working with the authors to investigate these concerns and establish what (if any) impact they will have on the paper's results and conclusions. In the meantime, readers are advised to use caution when using results reported therein.

 [Matters Arising](#) to this article was published on 25 August 2021

 A [Publisher Correction](#) to this article was published on 20 November 2020

 This article has been [updated](#)

《碳质硫氢化物的室温超导性》：其报道了碳质硫氢化物体系中的超导性，在 $267 \pm 10 \text{ kMPa}$ 下实现的最大超导转变温度为 15°C 。

室温超导前沿研究

[Submitted on 22 Jul 2023]

The First Room-Temperature Ambient-Pressure Superconductor

Sukbae Lee, Ji-Hoon Kim, Young-Wan Kwon

For the first time in the world, we succeeded in synthesizing the room-temperature superconductor ($T_c \geq 400$ K, 127°C) working at ambient pressure with a modified lead-apatite (LK-99) structure. The superconductivity of LK-99 is proved with the C magnetic field (H_c), and the Meissner effect. The superconductivity of LK-99 originates from by external factors such as temperature and pressure. The shrinkage is caused by Cu^{2+} subs and it generates the stress. It concurrently transfers to Pb(1) of the cylindrical column resulting in superconducting quantum wells (SQWs) in the interface. The heat capacity results indicated that LK-99. The unique structure of LK-99 that allows the minute distorted structure to be maintained maintains and exhibits superconductivity at room temperatures and ambient pressure.

Submission history

From: Young-Wan Kwon [[view email](#)]

[v1] Sat, 22 Jul 2023 07:51:19 UTC (2,620 KB)

[Submitted on 22 Jul 2023 (v1), last revised 11 Aug 2023 (this version, v3)]

Superconductor $\text{Pb}_{10-x}\text{Cu}_x(\text{PO}_4)_6\text{O}$ showing levitation at room temperature and atmospheric pressure and mechanism

Sukbae Lee, Jihoon Kim, Hyun-Tak Kim, Sungyeon Im, SooMin An, Keun Ho Aul

A material called LK-99, a modified-lead apatite crystal structure with the composition $\text{Pb}_{10-x}\text{Cu}_x(\text{PO}_4)_6\text{O}$ was synthesized by the solid state method. The material exhibits the Ohmic metal characteristic of $\text{Pb}_{6\text{S}1}$ above its T_c and the Meissner effect of a superconductor at room temperature and atmospheric pressure by the possibility of room-temperature superconductivity in this material is attributed to the insulator-metal transition achieved by substituting Pb with Cu, and the second being on-deformation in the one-dimensional(D) chain ($\text{Cu}^{2+}-\text{O}_{1/2}-\text{Cu}^{2+}$ along the c-axis) structure. The room-temperature T_c is discussed by 1-D BR-BCS theory.

Submission history

From: Hyuntak Kim [[view email](#)]

[v1] Sat, 22 Jul 2023 10:11:28 UTC (936 KB)

[v2] Sat, 29 Jul 2023 01:53:47 UTC (1,014 KB)

[v3] Fri, 11 Aug 2023 08:05:31 UTC (1,128 KB)

LK99: 研究显示这种材料中的杂质——尤其是硫化铜——才是电阻率快速下降并能部分悬浮于磁铁上方的原因，这些特性看起来与超导体很像。

5-3 Electromotive force and Kirchhoff's voltage law

Ohmic material under
static electric field

$$\oint_C \frac{1}{\sigma} \mathbf{J} \cdot d\boldsymbol{\ell} = 0.$$



Static electric field
is **conservative**

$$\oint_C \mathbf{E} \cdot d\boldsymbol{\ell} = 0.$$

*A **steady** current cannot be maintained in the same direction in a closed circuit by an electrostatic field. (D. K. Cheng, p. 206)*

*(**steady** 稳态: motion and constant velocity; **static** 静态: no motion)*

Electric fields inside an electric battery?

Impressed electric field, E_i

□ non-conservative electric field caused by chemical action;

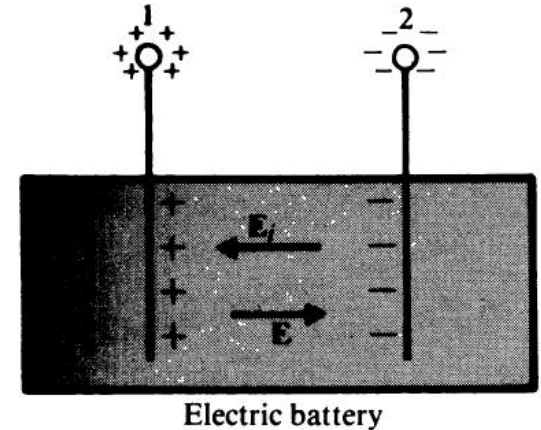
□ **Electromotive force (emf)** is the integral from the negative to the positive electrode **inside the battery**.

$$\mathcal{V} = \int_2^1 \mathbf{E}_i \cdot d\boldsymbol{\ell} = - \int_2^1 \mathbf{E} \cdot d\boldsymbol{\ell}.$$

The conservative electrostatic field satisfies

$$\oint_C \mathbf{E} \cdot d\boldsymbol{\ell} = \underbrace{\int_1^2 \mathbf{E} \cdot d\boldsymbol{\ell}}_{\text{Outside the source}} + \underbrace{\int_2^1 \mathbf{E} \cdot d\boldsymbol{\ell}}_{\text{Inside the source}} = 0.$$

emf: $\mathcal{V} = \underbrace{\int_1^2 \mathbf{E} \cdot d\boldsymbol{\ell}}_{\text{Outside the source}} = V_{12} = V_1 - V_2.$



Two different electric fields:

- (1) **Conservative fields** from the charges, both inside and outside the battery;
- (2) **Nonconservative fields-impressed fields**, from chemical reaction, only inside the battery

Connect the battery electrodes by resistance R

$$\mathbf{J} = \sigma(\mathbf{E} + \mathbf{E}_i),$$

Emf: (integral over the closed loop)

$$\mathcal{V} = \oint_C (\mathbf{E} + \mathbf{E}_i) \cdot d\boldsymbol{\ell} = \oint_C \frac{1}{\sigma} \mathbf{J} \cdot d\boldsymbol{\ell}.$$

- For one source of electromotive force
(if the internal resistance is negligible),

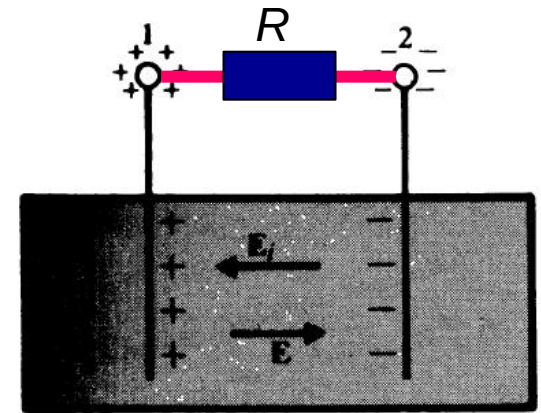
$$\mathcal{V} = RI.$$

- For more than one source of electromotive force and more than one resistor,

Kirchhoff's
voltage law

$$\sum_j \mathcal{V}_j = \sum_k R_k I_k \quad (\text{V}).$$

Around a closed path in an electric circuit, the algebraic sum of the emf's (voltage rises) is equal to the algebraic sum of the voltage drops across the resistances.



Electric battery

$$R = \frac{\ell}{\sigma S}.$$

$$J = I/S$$

5-4 Equation of continuity and Kirchhoff's current law

From the ***principle of conservative of charge***, the current leaving one region is the total outward flux of the current density through the enclosing surface S ,

$$I = \oint_S \mathbf{J} \cdot d\mathbf{s} = -\frac{dQ}{dt} = -\frac{d}{dt} \int_V \rho \, dv.$$

Apply divergence theorem, we obtain for a stationary volume

$$\int_V \nabla \cdot \mathbf{J} \, dv = -\int_V \frac{\partial \rho}{\partial t} \, dv. \quad (\rho \text{ is a time-space function})$$

The above equation holds regardless of V , and the integrands must be equal.

Equation of
continuity

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t} \quad (\text{A/m}^3).$$

For steady currents, $\partial\rho/\partial t = 0$.

$$\nabla \cdot \mathbf{J} = 0. \quad \leftarrow \quad \nabla \cdot \mathbf{J} = -\frac{\partial\rho}{\partial t}$$

Steady electric currents are divergenceless

$$\oint_S \mathbf{J} \cdot d\mathbf{s} = 0,$$

Kirchhoff's
current law

$$\sum_j I_j = 0 \quad (\text{A}).$$

The algebraic sum of all currents flowing out of a junction in an electric circuit is zero.

Relaxation time of charges in a metal?

Combining the Ohm's law with the continuity equation and assuming σ , we have

$$\sigma \nabla \cdot \mathbf{E} = -\frac{\partial \rho}{\partial t} \quad \leftarrow \quad \nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t}$$

In a simple medium, $\nabla \cdot \mathbf{E} = \rho/\epsilon$,

$$\frac{\partial \rho}{\partial t} + \frac{\sigma}{\epsilon} \rho = 0.$$

The solution is

$$\rho = \rho_0 e^{-(\sigma/\epsilon)t} \quad (\text{C/m}^3),$$

Relaxation time, τ

$$\tau = \frac{\epsilon}{\sigma} \quad (\text{s}).$$

Eg: Copper

$$\tau = 1.52 \times 10^{-19} \text{s}$$

The charge density at a given location will decrease with time exponentially.

5-5 Power dissipation and Joule's law

Power p provided by an electric field $\mathbf{E}$ in moving a charge q :

$$p = \lim_{\Delta t \rightarrow 0} \frac{\Delta w}{\Delta t} = q\mathbf{E} \cdot \mathbf{u},$$

Total power P delivered to all charge carriers in a volume dv :

$$dP = \sum_i p_i = \mathbf{E} \cdot \left(\underbrace{\sum_i N_i q_i \mathbf{u}_i}_{= \mathbf{J} \text{ (current density)}} \right) dv,$$

$$dP = \mathbf{E} \cdot \mathbf{J} dv$$

or

$$\frac{dP}{dv} = \underline{\mathbf{E} \cdot \mathbf{J}} \quad (\text{W/m}^3).$$

= power density

Total power converted into heat in a volume V

Joule's law

$$P = \int_V \mathbf{E} \cdot \mathbf{J} dv$$

In a conductor of a constant cross section, $dv=ds\bullet dl$, with dl measured in the current direction

$$P = \int_V \mathbf{E} \cdot \mathbf{J} dv$$



$$P = \int_L E d\ell \int_S J ds = VI,$$
$$P = I^2 R \quad (\text{W}).$$

$$p = \vec{J} \cdot \vec{E} = \frac{J^2}{\sigma} = \sigma E^2 \text{ is the microscopic form of Joule's law}$$

$$P = IU = I^2 R = \frac{U^2}{R} \text{ is the macroscopic form of Joule's law}$$

5-6 Boundary conditions for current density

In the absence of non-conservative energy source

Governing Equations for Steady Current Density	
Differential Form	Integral Form
$\nabla \cdot \mathbf{J} = 0$	$\oint_S \mathbf{J} \cdot d\mathbf{s} = 0$
$\nabla \times \left(\frac{\mathbf{J}}{\sigma} \right) = 0$	$\oint_C \frac{1}{\sigma} \mathbf{J} \cdot d\boldsymbol{\ell} = 0$

We can obtain the boundary conditions for $\mathbf{J}$
(as in Fig. 3-23 and in Sec. 3-9):

(1) Normal component of boundary current, J_n

$$\boxed{J_{1n} = J_{2n}} \quad \leftarrow \quad \nabla \cdot \mathbf{J} = 0$$

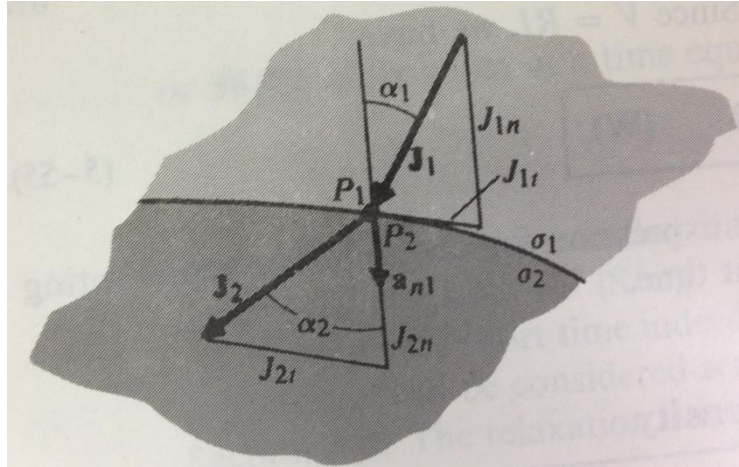
The normal components of J at two sides of an interface is continuous.

(2) Tangent component of boundary current, J_t

$$\boxed{\frac{J_{1t}}{J_{2t}} = \frac{\sigma_1}{\sigma_2}} \quad \leftarrow \quad \nabla \times (\mathbf{J}/\sigma) = 0$$

The ratio of the tangential components of J at two sides of an interface is equal to the ratio of the electric conductivities.

Q5-3 Two conducting media with conductivities σ_1 and σ_2 are separated by an interface. The steady current density in medium 1 at point P_1 has a magnitude J_1 and makes an angle α_1 with the normal. Determine the magnitude and direction of the current density at point P_2 in medium 2.



$$J_1 \cos \alpha_1 = J_2 \cos \alpha_2$$

$$\sigma_2 J_1 \sin \alpha_1 = \sigma_1 J_2 \sin \alpha_2$$

$$\frac{\tan \alpha_2}{\tan \alpha_1} = \frac{\sigma_2}{\sigma_1}$$

If medium 1 is a much better conductor than medium 2 $\sigma_1 \gg \sigma_2$ or $\frac{\sigma_2}{\sigma_1} \rightarrow 0$

$\alpha_2 \rightarrow$ zero, and J_2 emerges almost perpendicularly to the interface (normal to the surface of the good conductor). The magnitude of J_2 is

$$\begin{aligned} J_2 &= \sqrt{J_{2t}^2 + J_{2n}^2} = \sqrt{(J_2 \sin \alpha_2)^2 + (J_2 \cos \alpha_2)^2} \\ &= \left[\left(\frac{\sigma_2}{\sigma_1} J_1 \sin \alpha_1 \right)^2 + (J_1 \cos \alpha_1)^2 \right]^{1/2} = J_1 \left[\left(\frac{\sigma_2}{\sigma_1} \sin \alpha_1 \right)^2 + \cos^2 \alpha_1 \right]^{1/2} \end{aligned}$$

Boundary between two lossy dielectrics

When a steady current flows across the boundary between two different lossy dielectrics (dielectrics with permittivities ϵ_1 and ϵ_2 and finite conductivities σ_1 and σ_2),

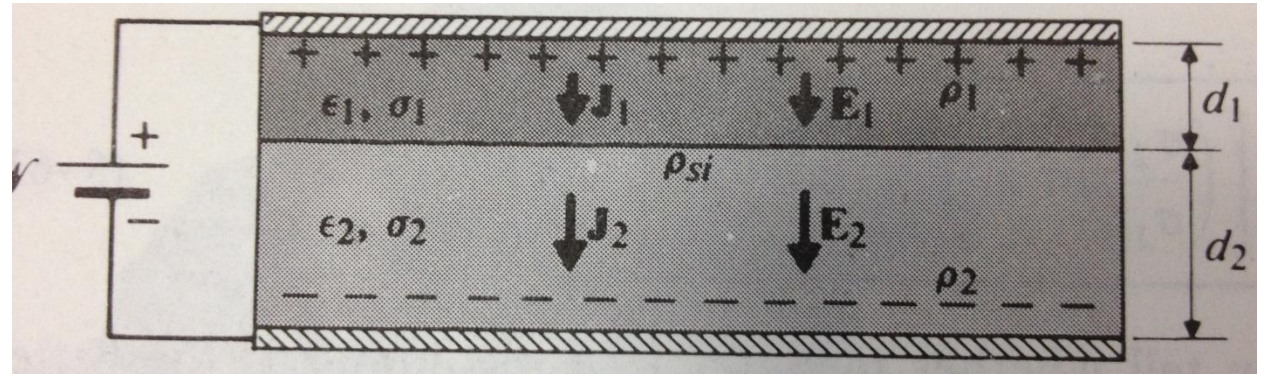
$$E_{2t} = E_{1t},$$

$$J_{1n} = J_{2n} \rightarrow \sigma_1 E_{1n} = \sigma_2 E_{2n}$$

$$D_{1n} - D_{2n} = \rho_s \rightarrow \epsilon_1 E_{1n} - \epsilon_2 E_{2n} = \rho_s,$$

$$\rho_s = \left(\epsilon_1 \frac{\sigma_2}{\sigma_1} - \epsilon_2 \right) E_{2n} = \left(\epsilon_1 - \epsilon_2 \frac{\sigma_1}{\sigma_2} \right) E_{1n}.$$


An emf γ is applied across a parallel-plate capacitor of area S . The space between the conducting plates is filled with two different lossy dielectrics of thicknesses d_1 and d_2 , permittivities ϵ_1 and ϵ_2 , and conductivities σ_1 and σ_2 , respectively. Determine (a) the current density between the plates, (b) the electric field intensities in both dielectrics, and (c) the surface charge densities on the plates and at the interface.



(a) $\vec{J}$ normal component
Kirchhoff's voltage law

$$\vec{J}_1 = \vec{J}_2$$

$$V = (R_1 + R_2) I = \left(\frac{d_1}{\sigma_1 S} + \frac{d_2}{\sigma_2 S} \right) I$$

Hence,

$$J = \frac{I}{S} = \frac{V}{\left(\frac{d_1}{\sigma_1} \right) + \left(\frac{d_2}{\sigma_2} \right)} = \frac{\sigma_1 \sigma_2 V}{\sigma_2 d_1 + \sigma_1 d_2} \quad (A/m^2)$$

(b) To determine E_1 and E_2 :

$$V = E_1 d_1 + E_2 d_2$$

$$J_1 = J_2 \quad \longrightarrow \quad \sigma_1 E_1 = \sigma_2 E_2$$

$$\longrightarrow \left\{ \begin{array}{l} E_1 = \frac{\sigma_2 V}{\sigma_2 d_1 + \sigma_1 d_2} \quad (\text{V/m}) \\ E_2 = \frac{\sigma_1 V}{\sigma_2 d_1 + \sigma_1 d_2} \quad (\text{V/m}) \end{array} \right.$$

(c) The surface charge densities on the upper and lower plates :

$$\rho_{s1} = \varepsilon_1 E_1 = \frac{\varepsilon_1 \sigma_2 V}{\sigma_2 d_1 + \sigma_1 d_2} \quad (\text{C/m}^2) \qquad \rho_{s2} = -\varepsilon_2 E_2 = -\frac{\varepsilon_2 \sigma_1 V}{\sigma_2 d_1 + \sigma_1 d_2} \quad (\text{C/m}^2)$$

The surface charge densities at the interface:

$$\rho_{si} = D_{2n} - D_{1n} = \varepsilon_2 E_{2n} - \varepsilon_1 E_{1n} = \frac{(\varepsilon_2 \sigma_1 - \varepsilon_1 \sigma_2) V}{\sigma_2 d_1 + \sigma_1 d_2} \quad (\text{C/m}^2)$$

From these results we see that $\rho_{s2} \neq -\rho_{s1}$ but that $\rho_{s2} + \rho_{s1} + \rho_{si} = 0$

Chapter 5 Steady Electric Currents

Current density

Continuity of equation

Power dissipation

Ohm's law

Joule's law

Boundary conditions for current density

Current density

$$\mathbf{J} = Nq\mathbf{u}$$

$$\mathbf{J} = \rho\mathbf{u} \quad (\text{A/m}^2),$$

Continuity of equation

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t} \quad (\text{A/m}^3).$$

Ohm's law

$$\mathbf{J} = \sigma\mathbf{E} \quad (\text{A/m}^2),$$

Joule's law

$$p = \vec{J} \cdot \vec{E} = \frac{J^2}{\sigma} = \sigma E^2$$

Boundary conditions for current density

Governing Equations for Steady Current Density	
Differential Form	Integral Form
$\nabla \cdot \mathbf{J} = 0$	$\oint_S \mathbf{J} \cdot d\mathbf{s} = 0$
$\nabla \times \left(\frac{\mathbf{J}}{\sigma}\right) = 0$	$\oint_C \frac{1}{\sigma} \mathbf{J} \cdot d\boldsymbol{\ell} = 0$

$$J_{1n} = J_{2n}$$

$$\frac{J_{1t}}{J_{2t}} = \frac{\sigma_1}{\sigma_2}$$

5-7 Resistance calculations

Conductor resistance

$$R = \frac{V}{I} = \frac{-\oint_L \vec{E} \times d\vec{l}}{\oint_S \vec{J} \times d\vec{S}} = \frac{-\oint_L \vec{E} \times d\vec{l}}{\oint_S \sigma \vec{E} \times d\vec{S}}$$

The procedure for computing the resistance of a piece of conducting material between specified equipotential surfaces (or terminals):

1. Choose an appropriate coordinate system.
2. Assume a potential difference V_0 between conductor terminals.
3. Find electric field intensity E within the conductor.

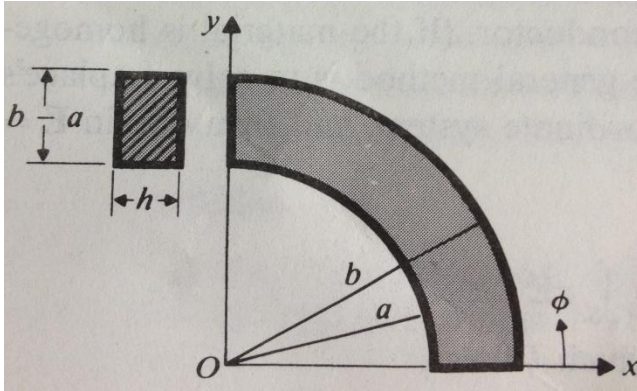
homogeneous material: Laplace's equation $\nabla^2 V = 0$ $\vec{E} = -\nabla V$

4. Find the total current

$$I = \oint_S \vec{J} \times d\vec{s} = \oint_S \sigma \vec{E} \times d\vec{s}$$

5. Find resistance R by taking the ratio V_0/I .

Q5-6 A conducting material of uniform thickness h and conductivity σ has the shape of a quarter of a flat circular washer, with inner radius a and outer radius b . Determine the resistance between the end faces.



Appropriate coordinate: cylindrical coordinate

Assume a potential difference V_0 between the end faces: $V=0$ on the end face at $y=0$

$V=V_0$ on the end face at $x=0$.

Solve Laplace's equation subject to the following boundary conditions

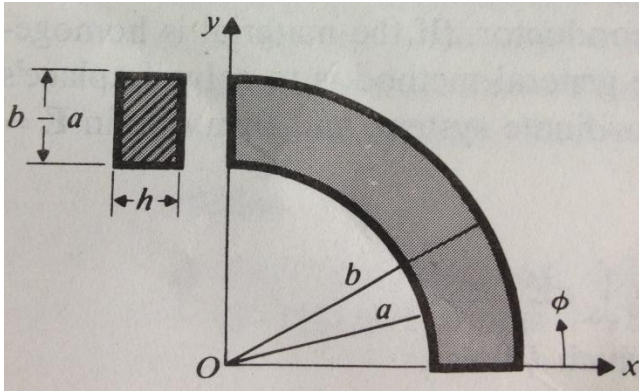
$V=0$ at $\Phi=0$

$V=V_0$ at $\Phi=\pi/2$

$$\frac{d^2V}{d\phi^2} = 0 \quad \longrightarrow \quad V = c_1\phi + c_2,$$

$$V = \frac{2V_0}{\pi} \phi$$

$$\nabla^2 V = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial V}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2}$$



The current density is

$$\vec{J} = s \vec{E} = -s \nabla V = -\vec{a}_r s \frac{\partial V}{r \partial f} = -\vec{a}_r \frac{2sV_0}{\rho r}$$

The total current I can be found by integrating $\vec{J}$ over the $\phi = \frac{\pi}{2}$ surface at which $-d\vec{s} = -\vec{a}_r h dr$

$$I = \oint_s \vec{J} \times d\vec{s} = \frac{2sV_0}{\rho} h \int_a^b \frac{dr}{r} = \frac{2shV_0}{\rho} \ln \frac{b}{a}$$

$$R = \frac{V_0}{I} = \frac{\pi}{2\sigma h \ln(b/a)}$$

Note that, for this problem, it is not convenient to begin by assuming a total current I because it is not obvious how $\vec{J}$ varies with r for a given I . Without $\vec{J}, \vec{E}$ and V_0 cannot be determined.